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Cyd Cipolla

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Build It Better: Tinkering in Feminist Maker Pedagogy

Cyd Cipolla

New York University, New York

“Taking responsibility for the social relations of science and technology means refusing an anti-science metaphysics.” —Donna Haraway, *The Cyborg Manifesto*

It is the first day of a new semester. I am perched on a couch in the reception area of the LaGuardia Co-Op, one of New York University’s shiny cooperative technology centers, trying not to pace. My students are all in the classroom at the other end of the building. I am generally veteran enough as a teacher to know how to hide my first-day jitters, but this is not just any first class meeting.

“How will we know if they did it?” the young man behind the front desk asks.

“We’ll know,” I say.

pedagogical trust

As if on cue, we hear the pinging tones of an 8-bit arrangement of the *Mario Brothers* theme, and, after somewhat of a confused pause, a loud cheer.

“Sounds like they did it,” I say, and head down the hallway to my classroom. There I find the members of my seminar-plus-lab course Tinkering in Feminist Technoscience huddled around an Arduino. The room bears the evidence of a frantic hour—chairs askew, backpacks open on the floor—as they have just completed their first day of lab by solving an interactive computer game. It began as a text adventure (a format with which none of my students are very familiar), and progressed, quite without warning, into a scavenger hunt, a pictographic puzzle, and then this final phase of electrical engineering. That day served as their crash course not only in how a text-based game works, but also in how to cooperatively explore the permeable boundary between simulation and reality, between text and material.

I love that students took this time to get to know each other on their own while acclimating to this new course environment

This first-day exercise models, in a fun and engaging way, a theme I repeat to my students often: our course is experimental in more ways than one. It is a standard twice-weekly humanities seminar attached to a coding and physical computing lab, and so, while I am asking the students to experiment (formally) with building, coding, and technology, and (informally) with the definitions of “maker,” “crafter,” and “engineer,” we are also all, together, experimenting with structures of pedagogy. Specifically, we are asking if it is possible to break down the boundaries between a classroom space, a maker space, and a laboratory space. This course— its proposal, evolution, and reception—forms the cornerstone of what I have come to call *feminist maker pedagogy*.

Generally, feminism is understood as a form of critical theory—a way of highlighting inequality and bias, particularly along lines of gender, sex, and sexuality. Introducing “making” as a pedagogical tool into a feminist classroom highlights the importance of moving beyond critique. **Feminist makers critique, but they also engage, speculate, and generate.** Feminist maker pedagogy is an interdisciplinary pedagogy in which the “learn by doing” model employed in laboratory courses is brought to a wider application: one that assumes that what one “learns by doing” is not just the task at hand, but something fundamental about how the world fits together and how we fit into it.¹ Together, participants consider the radical potential of building from scratch in the digital age, the ethical imperative to re-write the world around us, and the philosophical experience of tinkering with knowledge itself.

~ connections to speculative design

This article presents a rough backbone of feminist maker pedagogy, starting with its roots and related theories and then moving to reflections from my own practice. First, a definition: what I propose is *feminist* because it is concerned with the intersection of gender and sex with race, class, sexuality, and ability; it challenges neoliberalism, colonialism, and anthropocentrism; it pays attention to the politics of care and vulnerability; and it touches, at all points, on the embodied and interconnected experience of beings in living systems. What I propose is about *making* because participants in these exercises make material objects, even as they challenge the class, gender, and race dynamics at play in the contemporary maker movement. And finally, what I propose is a *pedagogy* because it describes a philosophy for teaching and learning, even as it challenges disciplinary boundaries, curricula, and the hierarchy of classroom spaces.

As implemented specifically in a technoscience studies course, the feminist maker pedagogy outlined here has three additional learning goals:

- (1) **The political potential of knowing how things work.** Students learn about everyday objects not just in an intellectual sense, but in a generative, critical sense. In a world where we usually access technology through simplified graphical interfaces, many students who have ideas for programs and objects to “make the world a better place” have no idea how those things are built, and, as a consequence, they cannot see how the underlying structures are unequal or inaccessible and thus how their projects might be furthering problematic structures of inequality. Transitioning from users to makers helps students understand the deeper issues. For example, I have had several students interested in the “Internet of things” (IoT)—objects that are connected to the Internet and can send and receive data. But as they deconstructed their ideas, sourcing parts and learning about how data are stored and sent, they began to ask questions

literacy that enables speculative designs that function otherwise

¹Similar to the argument Steve Mann makes in “Learning by Being.”

about privacy, cost, and surveillance. In the end, these students often significantly change their projects or abandon them altogether and emerge with a different perspective on IoT as a whole.

- (2) **The intellectual value of frustration.** To develop not only a tolerance for frustration, but the ability to see frustration as a useful and illuminating step on the way to understanding and discovery. Making demonstrates that sometimes in life you get directions that are unclear, incomplete, or inadequate, and that forging ahead anyway is not only rewarding but, in many cases, necessary—necessary for change, certainly, but for some, necessary for survival.

- (3) **The ability to apply these skills outside of the classroom.** The regular cross-pollination between lab and classroom ultimately improves learning and development because it will help students carry these tools over to their engagement in other interdisciplinary classroom spaces—that is, they may readily feel the importance of trial and error in trying to get their Arduino code to generate the right series of lights, and they will be asked to recognize the same feeling when they must return again to a complex section of a text in order to get it to work, so to speak. Productive employment of frustration and knowing the importance of how things work are helpful skills in both the theoretical and material worlds.

But first, why technology?

Just as feminist science studies engages a range of scientific and technical disciplines, and feminist perspectives inform the practice of a variety of scientific researches (Pollock et al.; Roy),² the laboratory portion of a feminist science studies course could include any number of scientific disciplines (George and Brenner; Giordano, “Building New”).³ I focus on programming and physical computing for practical reasons: ease of use, cost, availability, portability, a reasonable learning curve, and my own personal expertise. In addition to the support I provide in the lab itself, students have access to a robust online do-it-yourself (DIY)⁴ electronics community, which draws on a long history of home tinkering in radio, robotics, and engineering. They can also utilize university making, prototyping, and technology initiatives. Finally, students have a great deal of control over how difficult

²For an example of feminist, anticolonial science in a lab, see the Civic Laboratory for Environmental Action Research; <https://civillaboratory.nl/>.

³For an example, Clare Jen at Denison University teaches laboratory courses at the intersection of Women's Studies and Biology.

⁴Even when linked to explicitly radical politics, the term DIY and its attendant narratives run the risk of promoting individual solutions for structural problems, a persistent problem in underground movements. See Duncombe. Alternatives to DIY include DIT, *do-it-together* and DIO, *do-it-ourselves*. Although I use the term DIY in this article, note that my formation of feminist maker pedagogy distinguishes between self-reliance and isolationism and engages making as a cultural creative process that fosters connection. See Foster; Gauntlett.

there's something
affective to account
for here - an
affect of design?

transfer

I can also
see an
especially
timely
interest here -
more people
are interested
in small scale,
personalized,
and closed/
secure
technology that
is outside the
scope of
corporate technological
investment +
extraction

their projects are, allowing students with a wide variety of experience to take the same course.

Additionally, because of the perceived incongruity between feminist theory and computer science and engineering, there is a theoretical and pedagogical advantage to a feminist engagement with programming and physical computing specifically. Engineering and amateur electronics have long histories of exclusion (Hacker; Marvin), and the human-machine relationship, a system where “those who had access to certain machines were privileged, those who were expected to behave like certain machines were subjugated,” has always been structured along race, class, and gender lines (González 60). Feminist maker pedagogy that engages technoscientific building with feminist science and technology studies confronts this history head-on by making technology and feminist technoscience theory *mutually constitutive*. Students not only gain a feminist lens through technology but see technology as a formative, generative feminist tool.

In feminist maker pedagogy, we learn feminism through technology. We cultivate the idea that anyone can learn to build anything, given enough patience and a careful approach to documentation. Building their projects as a feminist engagement cements feminist theory as a grassroots, democratic, home-grown form of making and critiquing. Students engage with these forms of prototyping technology as a way of connecting to technology not as a cold, sterile, inaccessible process, but as something to be molded—paralleling the feminist invective that cultural roles are not things to be accepted, but systems anyone can understand, challenge, and change.

We also learn technology through feminism. Participants couple making with reading feminist studies of science and technology and computer science history and use them to understand themselves as engineers and technologists. By engaging feminism not as a secondary or tertiary form of critique, but as the foundational lens through which we examine technoscience, we directly confront the significant gender biases that exist within science, technology, engineering, and math (STEM; and particularly within computer science and engineering), break down the two-culture model of feminism versus science, and produce science out of feminist theory (Subramaniam and Willey).

Feminist science studies, politically, is deeply devoted to the democratization of science—to empowering amateurs, tinkerers, and critics alike to engage in the process of scientific knowledge formation. Feminist maker pedagogy builds on the principles of feminist science studies alongside those of feminist making and teaches that it is not enough to take things apart; you must also learn how to put them back together.

which models or feminist technoscience praxis
feminist technologists?

The roots of feminist maker pedagogy

Feminism as making, making as feminist

While the rest of this article traces the connections between feminist maker pedagogy and feminist science studies, I want to start, first, with connections to women's activism and feminism generally. Certain forms of material creation have long been associated with women, especially, and in some cases exclusively, when these activities are carried on inside the home and not for commercial purposes (Scott and Keates). Various scholars have examined the historical and contemporary role of craft-work as a culturally specific site of political agency, subversion, identity, and expression for people otherwise marginalized (Hackney, "Quiet Activism"; Hackney, "Use Your Hands"). This includes work on sewing, especially dressmaking, quilting, and embroidery (Burman; Clover; Gordon; Langellier; Parker); spinning and weaving (Creighton; McCafferty and McCafferty; Peskowitz; Rosner and Foster); and knitting (Greer, *Knitting for Good!*; Greer, *Craftivism*; Groeneveld; 2012; Turney).

In addition to the ways that "women's work" transforms into "women's activism" both historically and in the present moment, there are explicitly feminist movements associated with specific forms of crafting and making. Orton-Johnson, in her 2014 work on the rise of "craftivism," notes that it is "focused on the intersections of personal, social, and political engagement and third-wave feminism" (141). Red Chidgey notes how the rise of online idea- and pattern-sharing communities on sites like Tumblr, Pinterest, Etsy, and Ravelry incubate a form of DIY feminism, which is itself rooted in the 'zine culture of making in the '90s (Kempson; Piano). In fact, feminist crafting and DIY movements are more digital and technological than the division between crafting and making implies, and online crafting communities are often seen as tech-savvy (Von Busch) and tied to cyberfeminism (Bratich and Brush; Minahan and Cox).

More broadly, if we are to consider the idea that *making* means using resources available to you to create or construct something that solves a problem or addresses a need, then much of feminism is made up of makers. Building new things is an important part of feminist praxis, especially once we take into account feminism's long commitment to exploring lives on the ground. Feminist science studies has long been a community of makers, from Helen Longino's proposal that feminist science was not simply the elimination of bias but the construction of better frameworks ("Can There Be a Feminist Science?"), to Angela Willey's call for a queer feminist theory of biopossibility. And the wider feminist canon, however we may define it, is rife with scholars like Octavia Butler, Hélène Cixous, Mary Daly, Luce Irigaray, and Adrienne Rich who saw things in the world that did not yet have words to describe them, and so they made them up. And although all

irrigatingly,
old memory
chips
likewise
relied on
weaving
(cove rope
memory) -
this
practice
involved
women
weaving this
copper, including
for chips
used for
the Apollo
Guidance
computer



academic work to one extent or another serves to fill gaps in knowledge, feminist theorists prioritize novelty, applicability to human needs, and disruption of current paradigms (Longino, “Gender, Politics, and the Theoretical Virtues”) in ways that tie our work to the contemporary political stances taken by crafters and makers. In centralizing these connections between feminism and making, feminist maker pedagogy helps students to see the continued urgency of feminist work on the boundaries between public and private, personal and political, and domestic and commercial spheres. It also encourages them to apply their creative energy to finding new solutions to new problems, such as questions about how contemporary uses of social media and data collection intersect with and complicate these ongoing feminist concerns.

Critical maker identities

As the previous section highlighted with specific attention to feminism, making is not only a generative practice but a political one, and contemporary scholarship highlights the importance of a specifically critical making. Critical making has three intertwined themes: (1) reflection on the relationship between the maker and the thing being made; (2) focus not on the finished project but on the process of making itself, especially as done collectively; and (3) positing making as a form of intervention in hierarchical power systems (Ratto, “Critical Making”; Ratto, “Textual Doppelgangers”; Ratto and Boler; Wylie et al.).⁵

Despite the recent interventions represented by critical making, gendered, raced and classed assumptions about *maker* identity persist, especially when compared to other identities such as *craft*, *hacker*, or *engineer*.⁶ Women, queer people, and people of color, despite overwhelming interest in making and building, report feeling excluded by hacker and maker spaces in complex and covert ways (Davies). Claims about a new maker movement serve to reify these problems: it is only the case that there has been a resurgence in making if you are measuring the activities of upper-middle-class white people. If your definition of making includes crafting, sewing, growing, cooking, and fixing—if you consider not only “garage science” but also “kitchen science”—then you quickly see that people have been making continuously for centuries (Jen).

One way to challenge the gendered and raced lines around “making” and complicate the relationship between hard and soft engagements with

⁵As a practice, critical making is related to material thinking; some versions of hacktivism, especially as interpreted in maktivism, praxistemology, or tinkering; civic technoscience, craftivism, digital craft, and fabriculture; constructionism; critical technical practice; and critical, adversarial, and participatory design. There is also a direct relationship between articulations of critical making and engagements with democratic (and sometimes anarchic) forms of citizenship. See Ratto and Boler.

⁶See work on craftivism: Garber; Levine and Heimerl.

There is still aspects of Ratto's (2011) emphasis on critical making as a process of conceptualization that I am grappling with

and also, even when women do engage in this “garage” science, their work and contributions are subject to historical erasure + silencing

technology (Dunbar-Hester, “Beyond ‘Dudecore’”; Kafai and Peppler; Turkle and Papert) is to continue to assert that what crafters do *is* making and engineering. In this, we join the efforts of those in what Ellen K. Foster calls the margins of the maker community—feminist hacker collectives, library maker spaces, and fixer collectives—in troubling the mainstream categorization of the maker movement. Another way to challenge the dominant structure of making as innovation and entrepreneurship is that of “broken world thinking,” where we start from decay rather than progress (Jackson). Thus, one practice of feminist maker pedagogy is to assume that any space on a campus or in a city built for making or prototyping is a space for crafting, tinkering, and fixing—that the funds and resources made available for technology and entrepreneurship can, and should, be used to theorize materiality, to practice a “matter of care” (Foster), and to foster appreciation for the wonders of maintenance (Jackson.)

brings in scholarship of repair + maintenance connection to infrastructure

Assuming or asserting inclusion alone, however, is not enough. Dunbar-Hester argues, via Donna Haraway (*Simian, Cyborgs, and Women*), that the maker movement’s appeal to universality only appeals to those who feel universal to begin with—those whose identities are unmarked—and that activists should both consider the fact that making technology “has long been associated with white masculinity” and remember that it is not enough to simply express an embrace of egalitarianism (Dunbar-Hester, “Radical Inclusion?” 76). In that same article, Dunbar-Hester recommends using

→ is also being appended by...

Lucy Suchman’s “located accountability” to give learners a localized engagement point. An explicitly feminist maker pedagogy, and particularly one that is committed to decolonization, queering, antiracist, and anti-ableist agendas, both explicitly invites those who feel left out of a universal view and gives them the tools to name and challenge the hegemonic systems of exclusion. Dunbar-Hester’s observation rang true for my students, who often noted that they had no explicit reason to avoid maker spaces or technological engagement, only that they did not feel welcome. But many who did make their way to open-access spaces during our course reported that their initial feelings of trepidation were overcome by the knowledge that once they got what they needed (a 3-D print, a drill bit set, a power supply), they could come back to our lab to use it. And, for most of them, even the small introduction to critical technoscience in our class was enough to help them return to these spaces on their own, confident in their ability to belong.

→ the means to consider & articulate their situatedness in the context of technological production

Diversity in STEM

One of the main arguments for change in STEM education is inclusion—that we need to change the way we teach science in order to make it safer and more welcoming for women and people of color, with the aim of convincing more women and people of color to become scientists and engineers.

Suchman (2002)

Practitioners of feminist science pedagogy, myself included, find themselves caught between the imperative to bring more women into STEM and the imperative for a feminist interrogation of science, especially when the two are at odds (Mayberry; Subramaniam). Framing the lack of female representation among the highest professional levels of STEM fields as an issue of recruitment or retention leads to overly simple answers and misplaced blame, and this is particularly true in computer science and engineering. At best, and often openly, this framing places blame on the shoulders of those who teach and mentor young people, assuming “that girls avoid advanced computing because computer teachers continually say grossly sexist things to them in class, or because counselors urge them to take home economics instead, or because parents aggressively discourage their daughters from taking more than the minimum required” (Sanders 148). At worst, these efforts place blame on students for not being inspired or dedicated enough. Alison Adam notes that “this is the acceptable face of liberal feminism (Henwood) where the *status quo* is left unchallenged, where women constitute the problem for not entering computing in the numbers that they should” (Adam 157).⁷ Efforts to counteract the gender imbalances in STEM that do not engage the complex and systematic issues of sexism, racism, classism, and colonialism of scientific knowledge serve only to feed more vulnerable people into the leaky pipeline (Bilimoria and Liang).

In response, various scholars have argued for the transformation of science education through the incorporation of critical studies of science, and especially feminist science, into the natural science classroom (Barad, “A Feminist Approach”; Mayberry),⁸ or for greater cooperation between Women in Science Education (WISE) programs and feminist science and technology studies (Subramaniam). This tactic addresses not only feminist concerns about the place of situated and gendered knowledge (Hammonds; Haraway, “Situated Knowledges”; Harding), but also concerns about the misrepresentation of the formation of scientific knowledge as a monolithic, straightforward procession of facts “largely dependent on a dry process of ‘hypothesis testing’” (Nair and Majetich 26).⁹ As laudable, and as difficult, as this directive is, it is insufficient because it retains the idea that learning about science is just for (future) scientists. If we are to take seriously the dictum that feminist science is democratized science—*citizen science* (Giordano, “New Democratic Sciences”; Irwin)—then we must begin to directly challenge the separation of science student from non-science student. Feminist maker pedagogy, as I practice it, is aimed at making technoscience practices visible and accessible to all, but especially to those who do not

points to deeper paradigmatic issues within STEM disciplines which are also manifested in pedagogical inquiries

Problems of positivism, logics of objectivity, empiricism thus are also raced, sexed, gendered issues

⁷Internal quotation from Henwood.

⁸Similar to the final phase of a transformed biology curriculum described in Peggy McIntosh’s classic work on interactive phases of curriculum development (2).

⁹See also Barad, “A Feminist Approach to Teaching Quantum Physics.”

intend to pursue a career in the sciences. Or, put another way, where other articulations of feminist science pedagogy (Rose and Mayberry; Rosser) argue for bringing feminism into the science classroom, feminist maker pedagogy brings science and engineering into the feminist classroom. And where programs like WISE aim to remove boundaries so that students with interest and aptitude in science feel encouraged to pursue STEM careers, feminist maker pedagogy engages students who are not explicitly interested in science at all.

I have found, in my years of interdisciplinary liberal arts teaching, that some students work hard to resist engagement with the practice of science, and they are aided in this endeavor by university science programs that limit access to laboratory work to serious undergraduate science majors (often with the justification that laboratory time is precious and expensive). The result of this, however, is that university science for non-science students means lectures, slides, heavy textbooks, tests, and memorization—an exercise in the *absorption* of knowledge rather than its *production*. Thus, our most science-phobic students do not engage one of the central methods of scientific knowledge: iterative experimentation. By this I do not mean simply the process of properly forming and testing a hypothesis and reporting on the results, but the tiring, slow, long, and steady process of hours at the bench and the many failed attempts behind every published success. The result is not only a lack of critical science literacy but also a lack of practical experience with frustration, trial and error, and hands-on learning. Both deficits serve to deepen the divide between science and non-science students and to replicate the disciplinary boundaries we, as feminist science scholars, are trying to break down.

Greater collaboration between humanities, arts, and sciences requires work on both sides. While those who work and teach in STEM programs and engineer schools advocate for critical science studies and greater exposure to arts and humanities within the curriculum, I argue that those of us in arts and humanities programs should invite the practice of science into our science studies classrooms. We should do so even if, *especially* if, we are not formally trained in traditional scientific disciplines. I do not have any formal science training. I learned to code on my own in my 30s, and I am quite terrible at it, but I can make things work when I need to. I felt, at first, that because I did not have formal training, I could not properly teach feminist studies of science that incorporate technological making—despite my great personal interest in computer science and electrical engineering. But just as the explicit naming of the course as feminist situates it as a place for those critical of science in particular ways, my own outsider, novice, tinkerer status encourages participation and enrollment in a way that programs situated inside STEM department do not, regardless of diversity and inclusion efforts.

into the role of scientist, designer, programmer, technologist, or maker
placing boundaries on who is granted access to that identity

I relate to this as a scholar + instructor very deeply

Tinkering

Before moving on to the presentation of a case study in the feminist maker course Tinkering in Feminist Technoscience, I want to explain the origins and significance of its title. I chose tinkering as a central metaphor originally both because of its use in other areas of critical making and its casual overtones: to tinker is to “build something, first, and then figure out what it is or what it can be useful for afterward. It is possible that some of these things may never be useful in any traditional sense” (Mann, “Maktivism” 32). Despite the widespread efforts of the maker movement, most of my students saw engineering, making, building, or science as something that could be practiced only for profit or for a grade—not for fun. I wanted students to approach technology, first and foremost, as something that could be played at, for a moment. In this I drew primarily from Sara Giordano’s work interrogating the use of the word “tinker” within the maker and DIY movements (Giordano, “New Democratic Sciences”). Giordano’s most recent work argues for a feminist ethics of tinkering defined as playful, response-able, iterative praxis (Giordano, “Theorizing Feminist Tinkering”).

There are many creative and productive practices I have used in my explorations of feminist maker pedagogy, and in Tinkering in Feminist Technoscience, that I offer as answers to Giordano’s call for feminist tinkering: playing cooperative games (as we do the first day of class), building, dismantling, and fixing toys, learning from children’s science playsets, annotating and illuminating code (and here I mean with marginalia, not just with inline comments), self-ethnographically narrating the process of debugging, and using notebooks to record handwritten reflections to both the readings and the laboratory projects. Above all, the most immediately salient pedagogical practice has been the choice to grade students on process and reflection rather than results. I found that when there was no pressure to compete with each other or a deadline for a working prototype, students felt empowered to experiment not only with technology, but with *telos*: purpose, objective, goal.

What is the measure of tinkering success if not a “working” product (even though mine didn’t end up working how I desired)? If it’s not about a reward for a “good” product, what is the reward? Perhaps the feminist technoscientific endeavor not only changes the reward, but actually entirely scraps the system of reward-motivation that pushes toward a tangible goal as the marker of completion. Maybe the work will never be done, but we are motivated to do it anyway. Muñoz reminds us “queerness is not here [and] we are not yet queer.” In fact, “we may never touch queerness,” but queerness is also “a longing that propels us onward” (Muñoz). What would it mean to experiment with the full understanding that the results will always be complex and incomplete, which is precisely why we need to keep doing it?¹⁰

revisit this!
The notion
of “tinkering”
I would
describe as
incredibly
central to my
scholarship
identity

queer
futurity
&
speculation
or
potentiality

¹⁰This quotation and all following excerpted from student work with permission and under condition of anonymity. Internal quotations from Muñoz.

As I read their work interweaving technological tinkering and queer and feminist politics, I realized I had been asking my students to be tinkerers even before I introduced electronics and coding into my feminist studies classrooms. As Dunbar-Hester notes, “tinkering is as much a form of cultural production as a technical one,” one that produces “not just technical artifacts but egalitarian social relations by eroding boundaries between experts and laypeople” (Dunbar-Hester, “Radical Inclusion?” 75).¹¹ As I taught feminist theory and women’s studies, I expected only a small minority of my students would go on in life to be professional feminist theorists, yet I asked them, semester after semester, year after year, to play at it for a moment—to read, analyze, critique, research, and engage and, in doing so, to challenge hierarchies and understand systems of power. And yes, they did it for a final grade, but—and this is to me an essence of feminist pedagogy and the teaching of intersectional feminist and critical theories of identity—they also did it for life.

Feminist maker pedagogy in action

Course overview

Tinkering in Feminist Technoscience is a twenty-student, interdisciplinary seminar about feminist science and technology studies with an attached maker lab.¹² I originally designed the course as an experiment to see what it would mean to have a lab as part of a traditional humanities course, and as such, the course has two components: a twice-weekly discussion-based seminar, for which students read texts in feminist studies of science and technology, and a once-weekly lab section, where students spend the first third of the semester learning basic coding and electronics and then the remainder working on their own projects.¹³ Students are required to build a new technological object in one of three categories: a text-based game coded in Inform¹⁴; a wearable visual art project or interactive object using an Arduino microcontroller¹⁵; or an advanced project of their own design. Due to support from a departmental grant, students can build from a communal class supply of materials and do not have to pay for anything out of pocket. Students document their journeys in reading and building in laboratory notebooks, present their projects to the class, and write final papers in

¹¹For more on the radical potentials of tinkering with theory, see Higgins, Wallace, and Bazzul.

¹²All syllabi and a full course description are available on my website, cyclopolla.com.

¹³I have tried incorporating making into the classroom in various ways, including blending the seminar and the lab together, holding weekly optional drop-in lab hours, an intensive six-week summer course where we held lab each week, and holding single lab sessions within larger seminar courses.

¹⁴An introductory pseudo-coding language. See Inform7.org.

¹⁵Arduinos are programmable microcontrollers that allow you to build interactive objects and wearables. See Arduino.cc.

which they connect their own journeys as feminist technoscientists to class readings. In order to encourage students to focus on the process of making itself rather than the finished product, I base final grades not on how well the project works, or whether it works at all, but on the different questions they ask in order to finish it.

The course, as a whole, aims to implement a feminist maker pedagogy where the process of building is not a means to an end, but a metaphor for engaging theory—specifically, in this case, feminist theories on science and technology. Unlike a traditional science class, the lab is not a space for formally testing or demonstrating the facts we have learned in the classroom but rather an integrated space that encourages students to think differently:

As different frustrations emerged, such as sewing failures, the shaking of my hands when soldering, or the inevitable failure of the code ... I embraced that I was working within a feminist science realm, and tossed aside my conceptualizations of what would be deemed “traditional.” Unlike traditional science, feminist science valued novelty, ontological heterogeneity, and nonlinear notability. There was significance and importance in the disruptions of what I was producing, there could be a multitude of answers to a singular problem, and there did not always have to be a solution. [...] I allowed, as my professor instructed, these feminist ideas and theories to guide me through my building experience. I truly believe, as well, that if we did not coincide this programming struggle with these lessons, my understanding of what I was doing would be completely different. I was constantly forced to reflect on an interdisciplinary level, that allowed both my object and myself to change simultaneously.

Spending time building in a lab environment contextualizes some of the questions and ideas posed by practicing scientists who draw from their own lab work to think critically about feminist theory (Roy; Subramaniam). Asking the students to chronicle both their reading responses and their laboratory work in one notebook helped them to see both reading and building as experiences of trial and error, anchoring the idea that some problems require a period of frustration and confusion before they can be solved. Finally, building a new object while reading neo-materialist (Braidotti) and intra-active (Barad, *Meeting the Universe Halfway*) feminist engagements with science led students to see themselves and their objects as partners in learning, and, sometimes, co-conspirators in political action.

Implementation

Since this class is designed to be an introduction to technological experience for students who had little to no background, I use the initial lab periods to teach basics skills of physical computing. We begin, as discussed earlier, with an introduction to problem solving and the world of text-based games. During the second week students learn how to wire and program an

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Arduino to control light emitting diodes (LEDs), and then in the third week they learn how to make the LEDs react to the environment using light or sound sensors. I dedicate the fourth and final of these structured lab sections to small group tutorials in motors and robotics, sewing using conductive thread, soldering, and 3-D modeling and printing. The remaining lab periods are entirely devoted to students working independently, first exploring online tutorials in their area of interest and then designing and building their own projects. I am there each week to facilitate, answer questions, and occasionally offer small group lessons when larger issues emerge.

It may seem from this description that teaching such a class would require comprehensive knowledge about the programs and platforms students might use, especially if the instructor wants to encourage students to explore and incorporate new ideas in their projects. But the goal for lab was not to teach them everything an Arduino is capable of, but rather to give students enough knowledge about the Arduino platform that they could navigate available resources for project building. In addition to showing them online resources such as Adafruit Industries, Instructables, Hackster, and programming forums like Github and Stack Exchange, I helped students make appointments with staff at our university 3-D printing facility and various maker spaces, brought in experts to teach them, and took trips with them to local hardware and electronics hobby supply stores.

I designed these introductory classes and project options for students with a range of experience, but especially for those who identify as art or humanities scholars. The potential for creativity of text-based adventure games (“[The Interactive Fiction Archive](#)”) along with the plain English look of the coding language Inform immediately appealed to writing and literature students, helping them to see code not just as something anyone can do, but as something they, specifically, could do:

I sat staring at the computer screen, listening to the tapping of everyone else’s keyboards, and I thought to myself, as someone majoring in human sexuality and writing, how much I did not belong in that room. But as time went on, I started to understand that what I was looking at was a *language*, it was an interactive *story*, and if there was one thing I could understand more than code, it was how to write a story.

The lessons on using Arduinos to program interactive lights appeal to students with backgrounds in art, sewing, and crafting, who see potential in being able to program LEDs as part of visual art displays, hanging tapestries, or clothing and other wearable electronics. Finally, knowing some of the students had prior experience in coding and programming, I allowed students to propose their own projects of any form. These students all took advantage of the safe exploratory lab environment to push the boundaries of their own knowledge or experiment with new technology. I also asked these students to serve as guides in the initial lessons, and they

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were a great help to their classmates in explaining basic coding and electrical circuitry. Having students help each other early on set an expectation for collaboration and cooperation during lab sessions and modeled for everyone what a democratized science might look like—an atmosphere that carried through to the end of the semester.

Before teaching the course for the first time, I wondered how much an inexperienced student could be expected to build. Of the 43 students who have taken the course so far, seven students coded games in Inform; 13 built wearable or visual art projects using programmed LEDs; 19 built interactive objects that incorporated sound, capacitive touch, or other similar feedback mechanisms; and four proposed their own projects (DIY virtual reality goggles; contributing to an open-source home assistant artificial intelligence; deconstructing and repurposing an eStim device; and working with a Jewelbot). Of the 35 students who chose to work on projects using micro-controllers, 16 built projects by following step-by-step tutorials and 19 designed their own projects by remixing or modifying online tutorials. All 43 students finished the semester having built something, and of those, 30 had functional prototypes. It is a small sample size, and there was a lot of variation, so it is difficult to draw concrete conclusions, but here are some general observations: (1) students with no experience can be expected to build a lot, provided that they are given sufficient materials and encouragement, and (2) experience turns out to come from a variety of sources.

Outcomes: Responsibility and resourcefulness

As of now, the most significant outcome of my practice in feminist maker pedagogy has been how it fosters personal experimentation—that is, how the blending of feminist technoscience studies, critical making, and larger activist movements encouraged students to experiment with things that were personal to them. Many of the students designed final projects that were overtly feminist and political. This was not a requirement of the class—in fact, they were not even required to design their own projects at all—but, citing feminism's challenge to objectivity (Haraway; Harding) and call for accountability (Roy; Willey), these students took the maker lab as an opportunity to confront bias in technoscience where it stood. Final projects included queer art pieces, wearables emblazoned with intersectional feminist quotations, games exploring feminist theory and women's history, interactive sex toys, and sex education guides. Students then reflected on the process of building these projects as experiments in the boundaries of knowledge, as meditations on frustration, and as explorations of the (often contradictory) calls for responsible civic engagement and entrepreneurial innovation.

These forms of personal experimentation have also looped back into the practices of pedagogy itself. The most recent time I taught the course, several

I can see how students also become acquainted with the principles of design thinking that Thom (2021) describes

students combined the ethics of feminist science studies with their own urgent investments in eco-social justice and climate change, challenging themselves to build with as little waste as possible. Where traditional STEM students might not question paying lab fees or wonder where to obtain materials, and other students found the stress of sourcing technological supplies and knowledge to be its own feminist adventure, these students delighted in their scavenger abilities—building with what they had or what they could borrow. One student took great pride in a robot she built entirely of objects that she found in her apartment at 2 a.m., including a chassis made of reused cardboard and wheels made from empty spice containers. Urged on by these practices, other students seized immediately onto the creative possibilities of resourcefulness by integrating technology into maker practices with which they already had experience, such as painting, sculpting, sewing, weaving, embroidery or knitting, and by designing projects based on the skills and ideas to which they knew they had access or that they could relate to prior knowledge. The result was a feminist technoscience practice of resourceful salvage, grounded in theories of sustainability and ecofeminism and tied to feminist maker pedagogy.

ecological
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of making

Conclusion: Frustration, pride, joy, and critique

The process of designing and teaching this course, with all of its tinkering and frustrations, has been a source of great joy and creativity to me, as a feminist scholar, as an educator, and as an amateur engineer. The largest portion of this joy, by far, has been the insights of the students as they sat, precariously and, with a nod to Nina Lykke, monstrously, at the intersection of feminism, science and technology studies, and the practices of making.

love this
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I encouraged students to express and embrace frustration throughout the course, but crossing the boundary between frustration with theory and frustration with making turned out to be more complex than I had anticipated. When we opened class discussion to questions of frustration and feeling inadequate, students were reluctant to admit weakness in front of their peers—but only when it came to the realm of traditional science and engineering. Almost all of the students in the class found the questions of raced and gendered dynamics in science and technology particularly personal. They spoke often of how discouraged they had become with science education, or the lack of support they had had in life to pursue coding, programming, or electronics as a hobby. Many also expressed frustration about the pressure they felt to be perfect—that, as marginalized people, in order to deserve a spot in a lab or a course they had to outperform their non-marginalized peers.¹⁶ They were also perfectly comfortable talking about how

¹⁶My students were caught in what Banu Subramaniam calls the space between the Matildas (the forgotten women of science) and the Curies (the extraordinary women of science) (203).

hard the readings were that we did for class, even the readings that were about science or that were scientific papers. Yet students in both classes were extremely reluctant to share frustrations they were having with their technoscience projects—frustrations I was reading about, in great detail, in their lab notebooks.

The day in seminar when I finally succeeded in persuading them to talk about their frustrating lab experiments, each student assumed they were the only one who could not get their project to work right. Most comments that day started with, “I am sure I am the only one having this problem, but...” The stakes for admitting failure in the practice of technoscience were higher than those for other forms of knowledge. Whether this was because they were comfortable enough with their intellectual ability in other areas to admit failure or because the stakes around science are always higher, I cannot say. What I can say, though, is that the pedagogical practice of thinking about frustration productively, and of seeing connections between frustration with theory and with practice helped students not only to see the incongruity but also how their own silence around their struggles was part of a larger system of silences:

Although I spent every week engaging with texts that taught me how women are held to impossible standards in the fields of technology and science, are inhibited in everything they do, and are belittled with their accomplishments being diminished or stolen from them, I negated identification with that struggle. I did not want to accept that I was experiencing a microcosmic example of this pervasive issue. What I ultimately gleaned from my project was that the unconscious biases and cultural problems presented in class are so deeply ingrained in our culture that it can take an immersive practice within the field to fully comprehend how to unravel these issues, and problematize them. I had to have hands-on experience with [coding] to recognize how these very issues touch my own life.

Another possible interpretation for why students were reluctant to discuss frustration with coding and physical computing is that science, math, and technology had been a joyless slog for so long that they no longer registered their frustration with the practice of technoscience as unexpected or noteworthy. This is not surprising, of course, not only because of the issues with science education outlined earlier, but also because, overwhelmingly, the ability to take pleasure in one’s own technological prowess is culturally and socially reserved for those who identify as male (Dunbar-Hester, “Beyond ‘Dudecore’”; Dunbar-Hester, “Geeks”; Kleif and Faulkner; Oldenziel; Varma)—and most of my students do not.

Once we broke through the barriers of silence, exclusion, and frustration, making technology gave many of the students moments of great joy—more specifically, it gave them a sense of joy that *needed to be shared*:

I look forward to times I can work on the project and I want to share my progress with every living soul I encounter (and often do!)[.] It's an elation I can't contain alone; it demands relationality and expansion. When I first made my shift register do what I want (flash my Morse code individually in a sequence), I experienced such a rush; I couldn't sit still or stay quiet, I had to jump around and shout a little, but even that wasn't enough. I needed social elation, so I wandered the 5th floor until I found my friend Katherine and dragged her into the office to be excited with me.

Another student noted the power she felt after mastering a new technology (in this case 3-D printing), and how it spilled over into a need to share joy not only with building but with learning feminist science:

I showed the [project] to every person I encountered, and all of my friends responded with the same feedback—"This is the first time you 3D printed? This is amazing!" Getting such positive reassurance made my confidence completely skyrocket. The days following my 3D printing experience, I felt untouchable. I began to openly talk about the project, what led me to it, and my knowledge of theory in feminist technoscience to my peers.

And yet, as a long-time scholar of queerness and sexuality, I cannot help but note how much erasure can happen in moments of euphoria and joy. As another student noted, feelings like joy and curiosity about technology are reserved for those who occupy positions of power:

Why is it that white men are rewarded by society for huddling around work benches piled high with assorted wires and electrical components, but Ahmed Mohamed, the Muslim student from Texas who was detained after his teacher mistook his self-made clock for a bomb, is emotionally scarred for doing the same? ... Curiosity has become a privilege for the few. Those who are at the forefront of these issues have been prevented from creating their own solutions. The diminished capacity of subjugated individuals to apply their situated knowledge to hacking solutions for their community reflects the existence of educational and wider social structures that have set boundaries on their ability to conceive of innovative solutions, causing them to become dependent on the vision of people who don't have the capacity to understand their respective struggles.

The approach I took in *Tinkering in Feminist Technoscience* positions making and building as revolutionary acts in and of themselves, a position based in the assumption that those who are making and building have the freedom to engage in building technology without becoming the subjects of increased surveillance. Yet, and this is the crucial bit, by coming at the practice of making from a fundamentally critical angle—that is, by engaging science from feminism—our class remained a space where students could be critical of science while also enjoying its practice.

And, in the end, I believe this is the most salient contribution of feminist maker pedagogy to feminist science studies practice, particularly in moments when we are discussing science out of feminism ([Cipolla et al.](#); [Subramaniam](#)

and Willey). It demonstrates to students that engaging in practices of technology, even while we remain deeply skeptical of those practices, is not only possible but necessary. You can find joy and pleasure in building and doing things well, even while remaining wary of techno-utopianism. In fact, my experience with feminist maker pedagogy has shown that a deep distrust of technology's ability to save us all makes the joy of building it all the more poignant. The declarations of TechCrunch Disrupt hopefuls who believe they can change the world, once so saccharine and overwrought, become intimately and personally familiar. Theirs is the exquisite joy of innovation and creation—a feeling so profound that it wipes away all of the frustration that came before. Feminist maker pedagogy helps students to understand why, in the tumble and rush toward that joy, it becomes easy—too easy—to push heady things like ethics to the side, especially for those whose identities are unmarked. The process of making an idea manifest is already so fraught with frustration and pitfalls that when we add unanswerable questions to the pile it becomes paralyzing. Feminist maker pedagogy allows students to sit with this paralysis—to wallow in the frustration, to recognize it, to take their time to move beyond it, and to use their powerful critical minds to do so. Then, when they move into the wider world, they know how complex, how impossible, and yet how crucial it is to demand that we *build it better*.

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